



SupplyChain Copilot

Your GenAI Partner from RFQ to Delivery

'Automate, Negotiate, Optimize – All In One Flow'

Agenda

Problem Statement

- UserStory [Real World Scenario]
 - Stakeholder Needs: A 4W Perspective
-

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SupplyChain Copilot – Solution

- How it Addresses Key Challenges
- Key Benefits of the Solution

2

Agentic AI [Agentic Orchestration]

- Agents, Tasks & Tools
-

Demo

- Process Flow
 - Technical Architecture
-

3

Target Domains & Customer Segments

- Future Enhancements (Nova Act + Beyond)
 - Project Build Timeline
 - Domains, Fit and Target Segments
-

4

Problem Statement

- Requirements start in **email threads, meetings and scanned documents** across departments.
- Manually must **manually extract** SKUs, quantities, budgets, dates and addresses from text.
- RFQs are then sent manually to suppliers, who reply in inconsistent formats (email, PDF/Handwritten, CSV).
- Data re-entry into Salesforce or ERP systems leads to errors and delays.
- Logistics and finance decisions on supplier/route are **slow and opaque**; risks (weather/ports/news) show up **late**. Missing optimal routes or budgets.
- **Result:** fragmented communication → delayed RFQs → costly logistics, higher CO2 → weak audit trail

4W [Who We Are -The 4W Canvas]	Description
Who	A mid-size enterprise (example: TechFlow Industries) managing multi-supplier sourcing.
What	Struggling to unify unstructured email/voice/docs, supplier responses and logistics data.
Why	Manual, disconnected workflows slow down decisions and increase costs.
Where	Operating across multiple geographies with internal teams (Finance, Logistics, Procurement).

User Story - Real-World Scenario

At TechFlow Industries, a mid-sized global manufacturer, procurement doesn't start with a structured RFQ form.

Procurement
Format

Chain of
Internal
Emails

Scanned
Documents

Meetings,
Recorded
Calls &
Voicemails

How It All Begins - The Email Thread

"We have an urgent requirement for 450 BMS-IND-48V-200AH units."
"Budget ceiling is \$270,000; delivery by Jan 30."
"Here's the shipping address and PO number."
"Ensure ISO 9001 and UL listed certifications."

This email thread itself becomes the starting point for our system.

- Sarah (Procurement): "We need 450 battery management systems for our new Phoenix line."
- David (Finance): "Budget approved - \$270K max, must deliver by Jan 30."
- Jennifer (Logistics): "Here's the delivery address and PO reference."
- Sarah (again): "Include quality certifications; no delivery delays tolerated."

In short, it's a messy, cross-departmental thread that **contains all the key details** - but in free-form, natural text.

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From: jennifer.wong@techflow-industries.com
Subject: RE: Urgent RFQ - Industrial Battery Systems Q1 2025
Date: October 16, 2025

Team,

For vendor coordination and delivery logistics:

Shipping Address:
TechFlow Industries Manufacturing
2847 Industrial Parkway, Building C
Phoenix, AZ 85034
United States

From: sarah.martinez@techflow-industries.com
Subject: RE: Urgent RFQ - Industrial Battery Systems Q1 2025
Date: October 16, 2025

Perfect coordination everyone. Let me provide final details for the RFQ package:

Company Information:
- TechFlow Industries (we're in the industrial automation and manufacturing sector)
- Corporate website: <https://www.techflow-industries.com/>
- Main contact number: (602) 555-0100
- Industry focus: Industrial automation systems and smart manufacturing

Special Notes: This is mission-critical for Q1 production launch. Zero tolerance for delivery delays. Quality certifications required (ISO 9001, UL listed).

Let's get this RFQ distributed to our approved vendor list today.

Best regards,
Sarah Martinez
Senior Procurement Manager
TechFlow Industries
2847 Industrial Parkway
Phoenix, AZ 85034
Direct: (602) 555-0123
sarah.martinez@techflow-industries.com

This is going to our Phoenix facility

Please use purchase order reference

Jennifer Wong
Logistics Coordinator
TechFlow Industries
2847 Industrial Parkway
Phoenix, AZ 85034
Phone: (602) 555-0123

From: david.kim@techflow-industries.com
Subject: RE: Urgent RFQ - Industrial Battery Systems Q1 2025
Date: October 16, 2025

Sarah,

Finance department approved the battery procurement. Budget ceiling is \$270,000 for roughly \$600 per unit maximum.

This is CRITICAL priority - the production line installation is scheduled and we cannot afford delivery no later than January 30th, 2025.

David Kim
Finance Director
TechFlow Industries
david.kim@techflow-industries.com

From: sarah.martinez@techflow-industries.com
Subject: RE: Urgent RFQ - Industrial Battery Systems Q1 2025
Date: October 16, 2025

Hi Procurement Team,

We have an urgent requirement for our Phoenix manufacturing management systems for the new production line expansion.

The engineering specs are BMS-IND-48V-200AH units. We need deployment.

Sarah Martinez
Senior Procurement Manager
TechFlow Industries
Phone: (602) 555-0123
sarah.martinez@techflow-industries.com

Our Solution – The GenAI SupplyChain Copilot

A **multi-agent GenAI orchestration system** on **Amazon Bedrock** with **Strands Agents**, that automates the **RFQ-to-Fulfillment** lifecycle.

- Understands emails, documents and voice inputs -> Generates structured RFQs, selects suppliers, negotiates, and plans logistics -> Delivers complete, autonomous, end-to-end procurement summaries
- This isn't automation; it's **cognitive collaboration**. It doesn't replace existing systems – it amplifies them.

End-to-End Workflow (Agentic Orchestration)

1. **Understand** internal email threads → create RFQ
2. **Validate** and update Salesforce (RFQ & Account)
3. **Evaluate** supplier responses and benchmark prices
4. **Negotiate** optimized pricing & terms
5. **Plan** optimal shipping routes (cost, ETA, CO₂)
6. **Summarize** full workflow with audit logs & Slack notifications

❑ Key Benefits

1

Reduction in cycle time from internal to supplier decision

2

Real-time orchestration of multi-modal data

3

Cross-system harmony (Salesforce, Slack, Email)

4

Auditability + Explainability built in

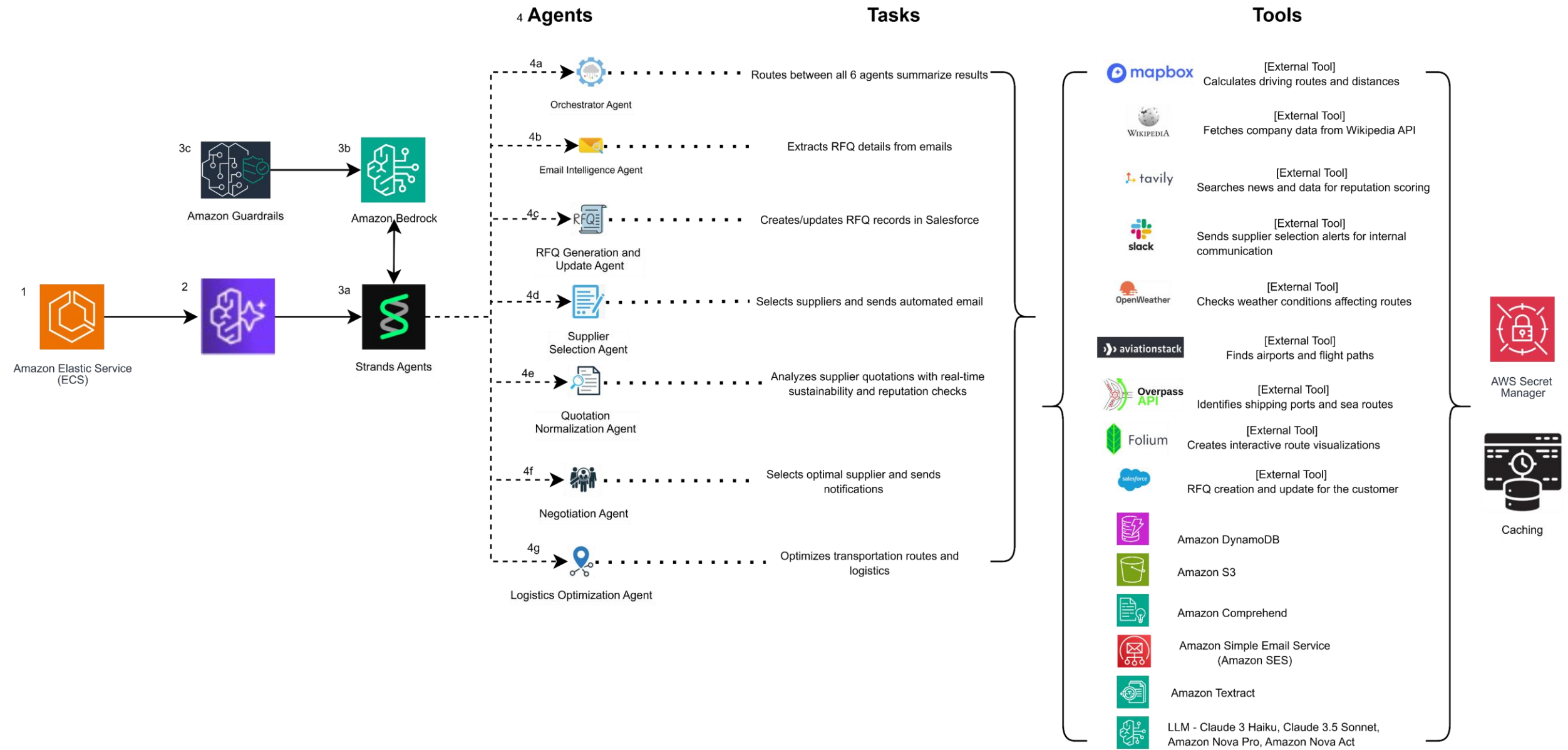
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Zero replacement: enhances existing supply chain stack

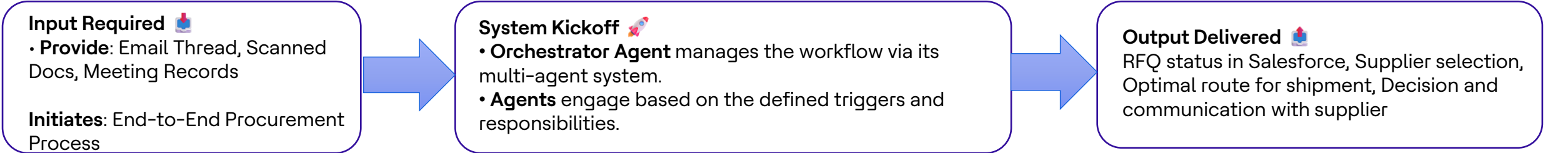
Agentic Orchestration [The Agents (0–5 + Supervisor)]

Agent	Role	Description	Key Tools & Usage
A0 – Email Intelligence Agent	 Reader	Extracts structured RFQ data from email threads, document attachments (PDF/handwritten) and voice transcripts	• Textract, Comprehend – Parse and extract entities • Gmail / SES – Fetch and process email chains • Claude 3.5 Haiku – Context-aware data extraction
A1 – RFQ Generation and Update Agent	 Recorder	Validates and updates RFQ records in Salesforce and DynamoDB	• Salesforce API – RFQ and Account updates • DynamoDB – Store RFQ and audit data • Claude 3.5 Haiku – Field mapping and validation
A2 – Supplier Selection Agent	 Evaluator	Scores suppliers by cost, lead time, and quality	• DynamoDB – Supplier data reference • Claude Haiku 3.5 – Multi-criteria evaluation • Tavily / Wikipedia – Reputation insights • SES / Slack – Outreach and notifications
A3 – Quotation Normalization Agent	 Analyzer	Normalizes supplier quotes and benchmarks market data	• Textract, DynamoDB – Quote extraction and Storage • Claude 3.5 Sonnet – Normalize quote formats • S3 – Quote repository
A4 – Negotiation Agent	 Communicator	Crafts counter-offers and manages supplier communications	• SES, Slack – Send negotiation updates • Comprehend – Sentiment and tone analysis • Claude Sonnet 3.5 Sonnet – Drafts optimized responses • Salesforce – RFQ data for negotiation
A5 – Logistics Optimization Agent	 Planner	Determines optimal shipping route and mode	• Mapbox / Folium – Route visualization • AviationStack, OpenWeather, Overpass – Route and weather data • Claude 3.5 Sonnet – Cost, ETA, and CO₂ optimization • DynamoDB – Route logs and metrics
A6 – Supervisor / Orchestrator Agent	 Orchestrator	Coordinates agent workflows and compiles final reports	• Amazon Nova Pro – Core orchestration and reasoning • AgentCore Runtime (ECS Fargate) – Workflow execution • CloudWatch – Monitoring and observability • DynamoDB, Slack – Audit and summary updates

Agentic Orchestration



Demo



❏ Demo Scenario

Email Conversation Intake (Internal → RFQ)

- Multi-departmental email thread (Procurement, Finance, and Logistics)
- Agent0 and Agent1 extract relevant data and creates/updates RFQ in Salesforce

1

Normalization & Selection

- Agent2 selects and ranks suppliers
- Quotation request sent through email to the selected supplier
- supplier replies quotes (PDF, CSV, Image)
- Agent3 extracts and normalizes quotes via OCR

2

Real-time data Analysis and Negotiation

- Agent4 receives RFQ and quote data
- Analyzes supplier reputation and sustainability via Google search
- Compares suppliers using weighted criteria to select the optimal one

3

Logistic Optimization

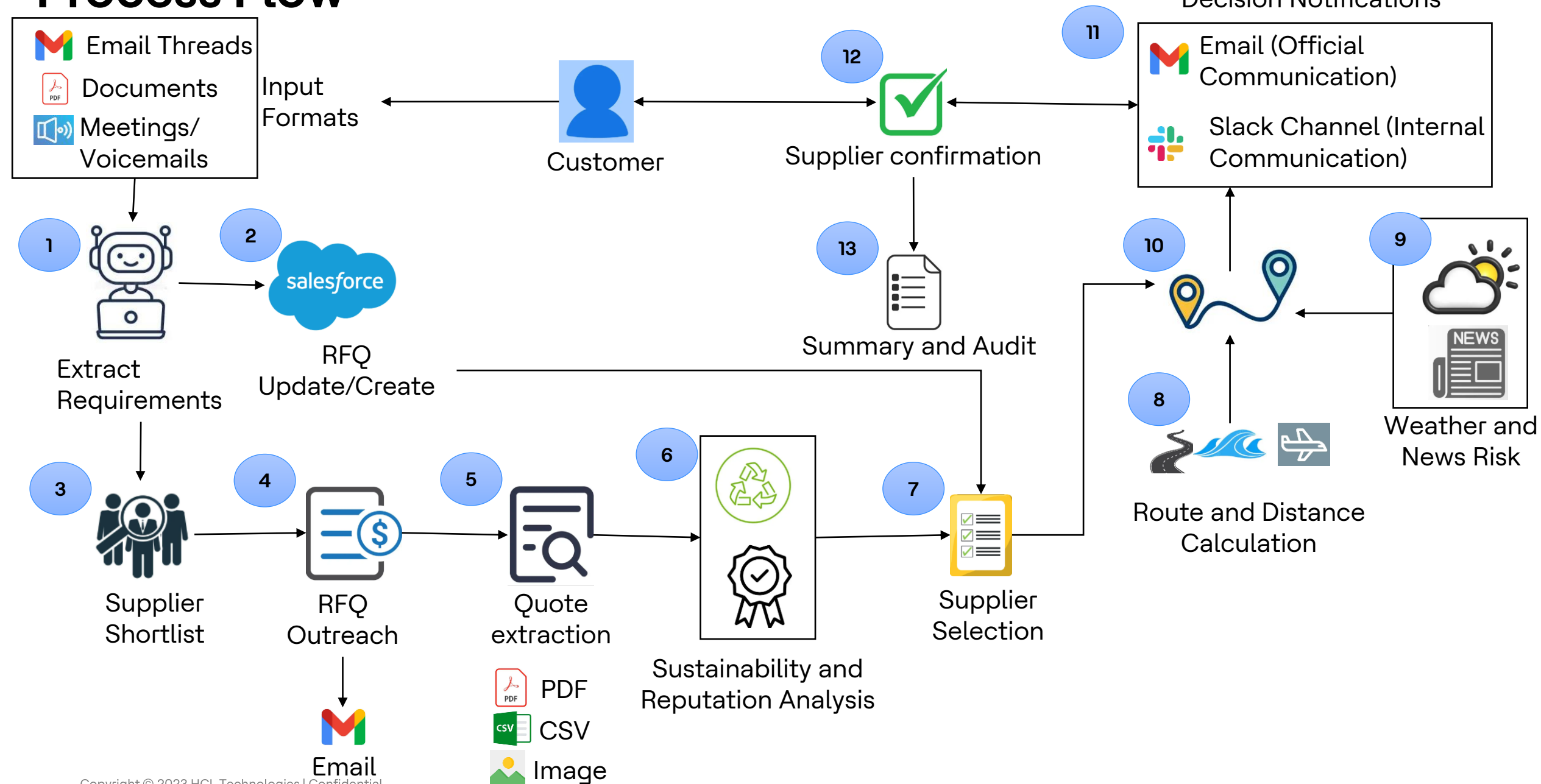
- Agent5 identifies shipment origin and destination
- Analyzes road, sea, and air routes with other metrics
- Maps optimal global routes for visualization
- Evaluates 7-day news and weather data to select the best shipping path

4

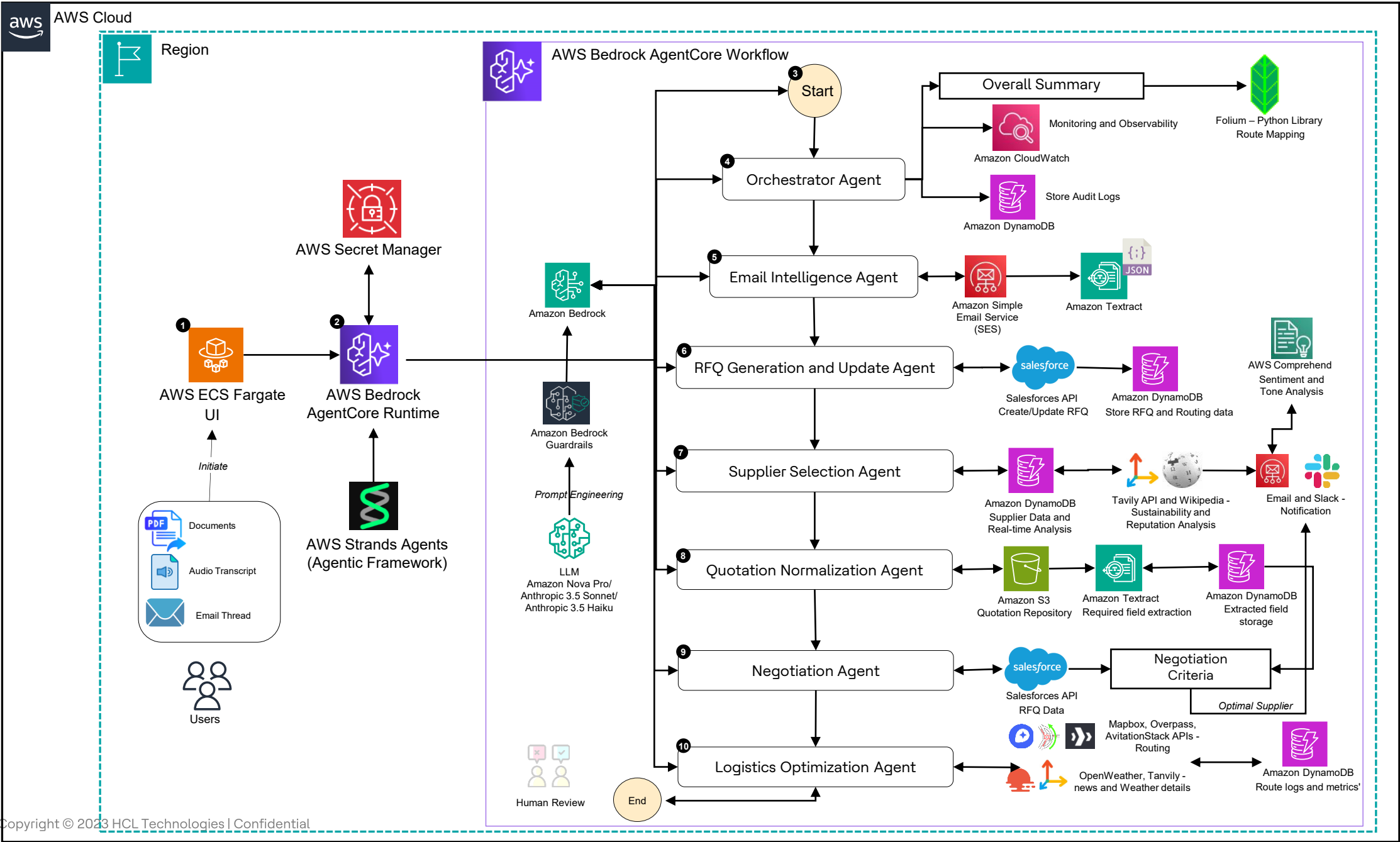
Demo

Video:

Process Flow



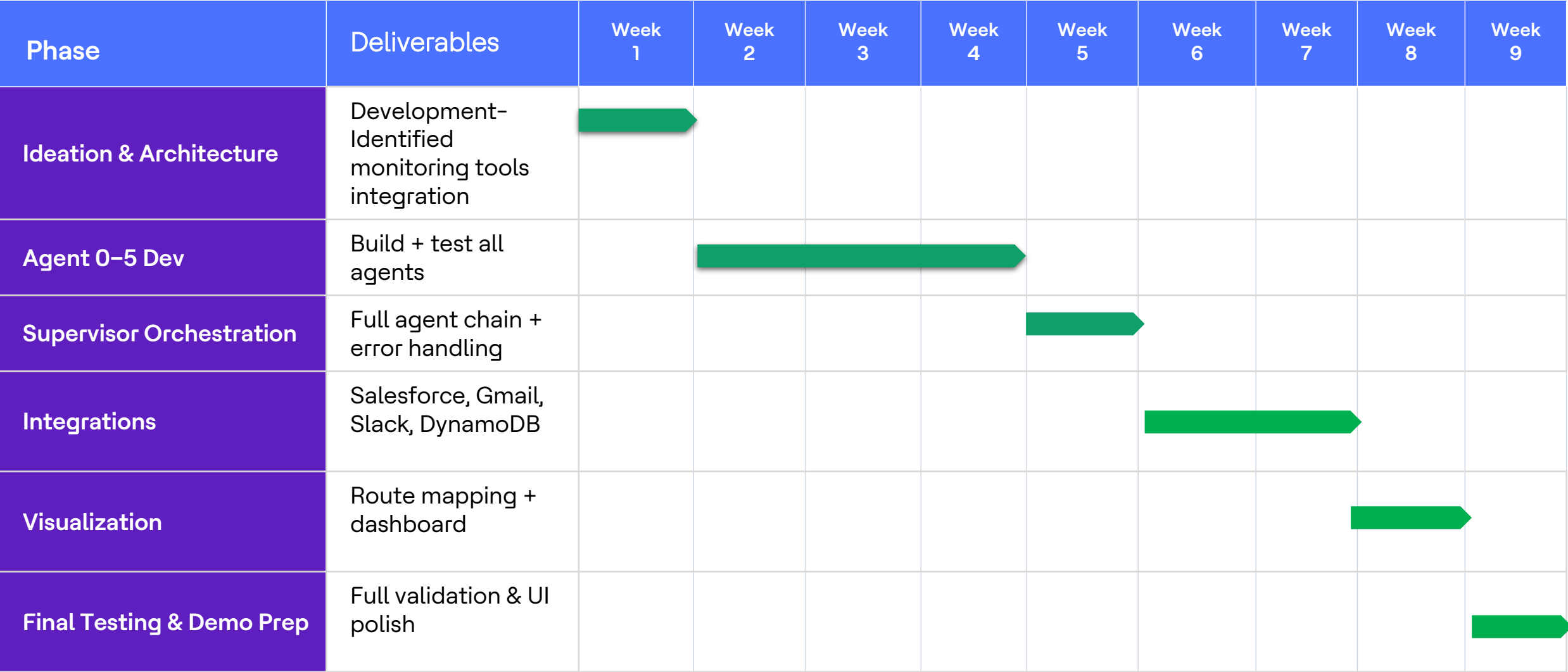
Technical Architecture



Future Enhancements (Nova Act + Beyond)

Enhancement	Description
Amazon Nova Act	Browser-action LLM to autonomously browse supplier sites and place orders
Guardrails Integration	Ensure safe, compliant outbound comms
RAG on internal SOPs	Use historical POs, QA docs for decision grounding
Predictive Disruption Engine	Anticipate port/weather/strike impacts
Caching Layer	Faster repeat quote/route lookups
Voice RFQs	Audio → Text → Email Intelligence Agent pipeline
Amazon Q Business Integration	Enable natural language Q&A over RFQ history

Project Build Timeline



Total : 8-9 Weeks

Production-ready prototype

Domains, Fit & Target Segments

Manufacturing & Industrial Automation

Multi-supplier sourcing; faster RFQs and disruption alerts.
(Procurement & Ops Teams)

Energy & Utilities

High-value assets; transparent RFQ tracking, smart logistics.
(Project Procurement Teams)

Aerospace & Defense

Complex sourcing; automated compliance checks.
(Strategic Sourcing & Compliance)

Electronics & Semiconductors

Global vendors, tight timelines; automated quote analysis.
(Category & Procurement Analysts)

Retail & Consumer Goods

Seasonal demand; real-time benchmarking and fulfillment.
(Category Buyers & Inventory Managers)

Construction & Infrastructure

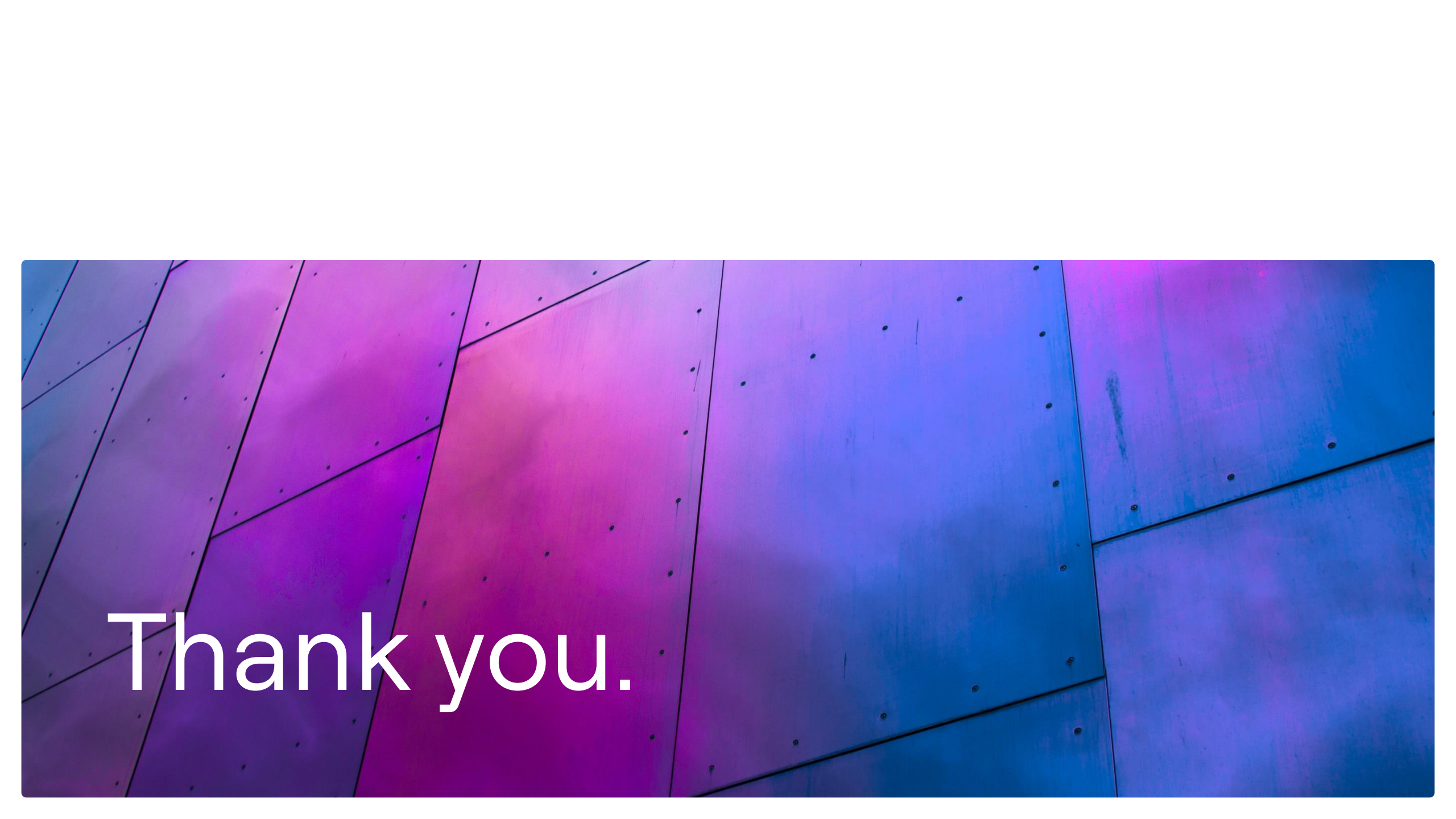
Frequent RFQs; cost-optimized material routing.
(Procurement & Project Managers)

Automotive & EV

Tiered suppliers; cost and CO₂-optimized sourcing.
(Procurement & Logistics Leads)

Pharma & Healthcare

Regulated, time-sensitive deliveries; optimized cold-chain routes.
(Supply Chain & QA Teams)

The background of the slide is an abstract composition of diagonal lines in various shades of purple and blue. The lines intersect to form a grid-like pattern of rectangular and trapezoidal shapes. The colors transition from a deep purple on the left to a vibrant blue on the right, with some areas showing a gradient. Small, dark, circular specks are scattered across the surface, resembling rivets or dust. The overall effect is modern and geometric.

Thank you.

